Patrick Foulis
CS457 Computer Networks
Lab 2
September 6, 2026

My Information

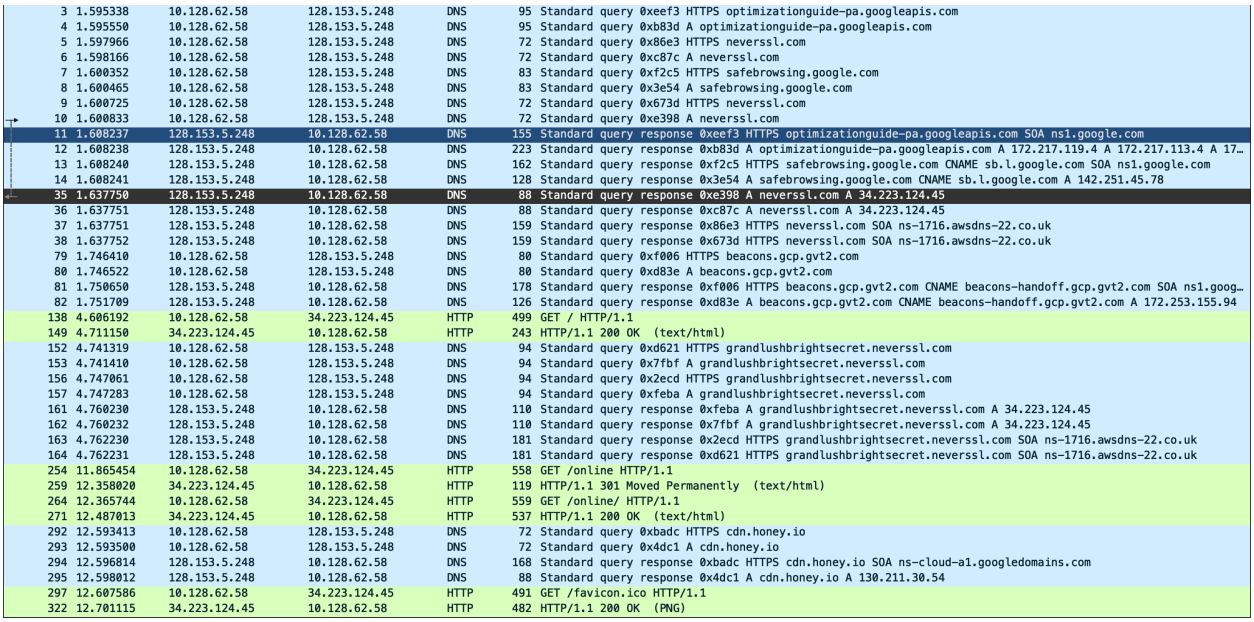
I am using a Mac with macOS Tahoe 26.6.2. I am not using WSL as I am on a Mac, and captured on en0 with Wireshark, which is the default wifi selection.

My First Capture

To get this capture I had to use the http://neverssl.com link because the example.com was redirecting me to https. I saved the file as lab2-http.pcapng located in the .zip file that I will attach. This captured roughly 331 packets in the loading process of the page.

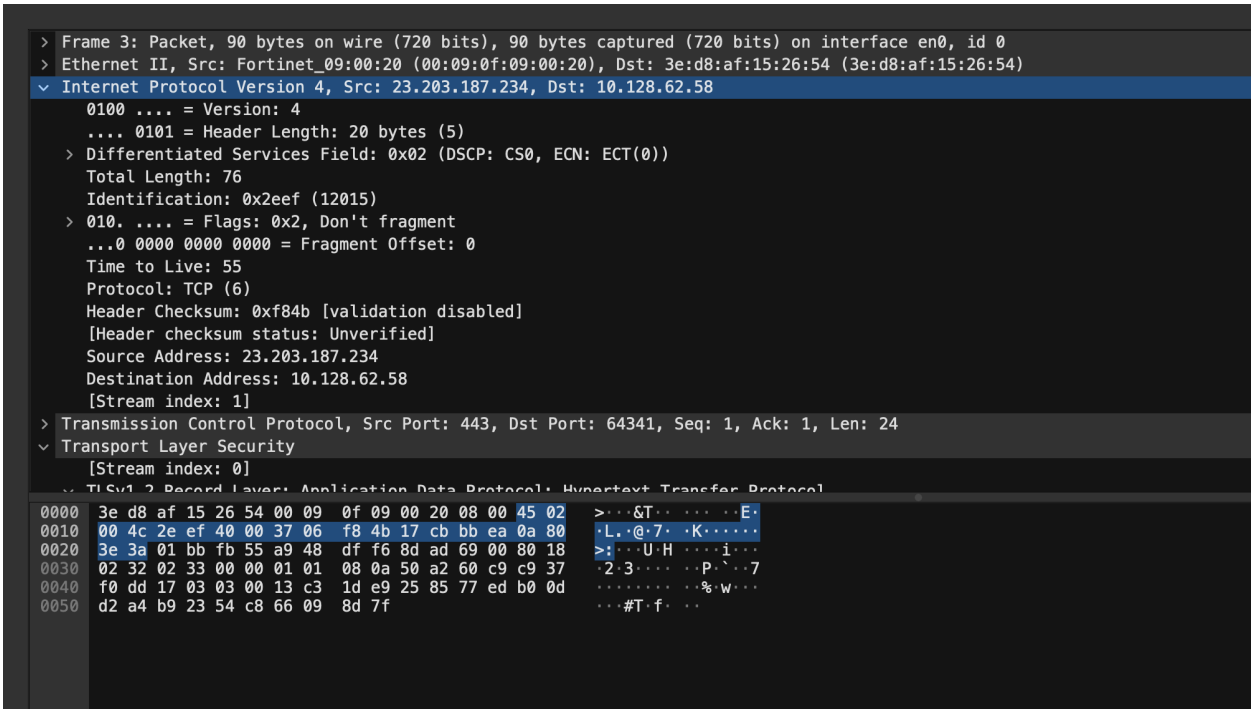
Display Sequence

In this capture, I was able to identify the domain name being queried. The root domain that was being queried was neverssl.com. There were a few other queries made but they were to get to the root query of neverssl.com. The IP address that was queried in this capture is, 34.223.124.45, this was shown in packets 35 and 36. The packet numbers for the GET was 138, 254 , 264, 297. The first three of these was for the HTTP request, and the subsequent ones were



3	1.595338	10.128.62.58	128.153.5.248	DNS	95	Standard query 0xeef3 HTTPS optimizationguide-pa.googleapis.com
4	1.595550	10.128.62.58	128.153.5.248	DNS	95	Standard query 0xb83d A optimizationguide-pa.googleapis.com
5	1.597966	10.128.62.58	128.153.5.248	DNS	72	Standard query 0x86e3 HTTPS neverssl.com
6	1.598166	10.128.62.58	128.153.5.248	DNS	72	Standard query 0xc87c A neverssl.com
7	1.600352	10.128.62.58	128.153.5.248	DNS	83	Standard query 0xf2c5 HTTPS safebrowsing.google.com
8	1.600465	10.128.62.58	128.153.5.248	DNS	83	Standard query 0x3e54 A safebrowsing.google.com
9	1.600725	10.128.62.58	128.153.5.248	DNS	72	Standard query 0x673d HTTPS neverssl.com
10	1.600833	10.128.62.58	128.153.5.248	DNS	72	Standard query 0xe398 A neverssl.com
11	1.608237	128.153.5.248	10.128.62.58	DNS	155	Standard query response 0xeef3 HTTPS optimizationguide-pa.googleapis.com SOA ns1.google.com
12	1.608238	128.153.5.248	10.128.62.58	DNS	223	Standard query response 0xb83d A optimizationguide-pa.googleapis.com A 172.217.119.4 A 172.217.113.4 A 17…
13	1.608240	128.153.5.248	10.128.62.58	DNS	162	Standard query response 0xf2c5 HTTPS safebrowsing.google.com CNAME sb.l.google.com SOA ns1.google.com
14	1.608241	128.153.5.248	10.128.62.58	DNS	128	Standard query response 0x3e54 A safebrowsing.google.com CNAME sb.l.google.com A 142.251.45.78
35	1.637750	128.153.5.248	10.128.62.58	DNS	88	Standard query response 0xe398 A neverssl.com A 34.223.124.45
36	1.637751	128.153.5.248	10.128.62.58	DNS	88	Standard query response 0xc87c A neverssl.com A 34.223.124.45
37	1.637751	128.153.5.248	10.128.62.58	DNS	159	Standard query response 0x86e3 HTTPS neverssl.com SOA ns-1716.awsdns-22.co.uk
38	1.637752	128.153.5.248	10.128.62.58	DNS	159	Standard query response 0x673d HTTPS neverssl.com SOA ns-1716.awsdns-22.co.uk
79	1.746410	10.128.62.58	128.153.5.248	DNS	80	Standard query 0xf006 HTTPS beacons.gcp.gvt2.com
80	1.746522	10.128.62.58	128.153.5.248	DNS	80	Standard query 0xd83e A beacons.gcp.gvt2.com
81	1.750650	128.153.5.248	10.128.62.58	DNS	178	Standard query response 0xf006 HTTPS beacons.gcp.gvt2.com CNAME beacons-handoff.gcp.gvt2.com SOA ns1.goog…
82	1.751709	128.153.5.248	10.128.62.58	DNS	126	Standard query response 0xd83e A beacons.gcp.gvt2.com CNAME beacons-handoff.gcp.gvt2.com A 172.253.155.94
138	4.606192	10.128.62.58	34.223.124.45	HTTP	499	GET / HTTP/1.1
149	4.711150	34.223.124.45	10.128.62.58	HTTP	243	HTTP/1.1 200 OK (text/html)
152	4.741319	10.128.62.58	128.153.5.248	DNS	94	Standard query 0xd621 HTTPS grandlushbrightsecret.neverssl.com
153	4.741410	10.128.62.58	128.153.5.248	DNS	94	Standard query 0x7fbf A grandlushbrightsecret.neverssl.com
156	4.747061	10.128.62.58	128.153.5.248	DNS	94	Standard query 0x2ecd HTTPS grandlushbrightsecret.neverssl.com
157	4.747283	10.128.62.58	128.153.5.248	DNS	94	Standard query 0xfeba A grandlushbrightsecret.neverssl.com
161	4.760230	128.153.5.248	10.128.62.58	DNS	110	Standard query response 0xfeba A grandlushbrightsecret.neverssl.com A 34.223.124.45
162	4.760232	128.153.5.248	10.128.62.58	DNS	110	Standard query response 0x7fbf A grandlushbrightsecret.neverssl.com A 34.223.124.45
163	4.762230	128.153.5.248	10.128.62.58	DNS	181	Standard query response 0x2ecd HTTPS grandlushbrightsecret.neverssl.com SOA ns-1716.awsdns-22.co.uk
164	4.762231	128.153.5.248	10.128.62.58	DNS	181	Standard query response 0xd621 HTTPS grandlushbrightsecret.neverssl.com SOA ns-1716.awsdns-22.co.uk
254	11.865454	10.128.62.58	34.223.124.45	HTTP	558	GET /online HTTP/1.1
259	12.358020	34.223.124.45	10.128.62.58	HTTP	119	HTTP/1.1 301 Moved Permanently (text/html)
264	12.365744	10.128.62.58	34.223.124.45	HTTP	559	GET /online/ HTTP/1.1
271	12.487013	34.223.124.45	10.128.62.58	HTTP	537	HTTP/1.1 200 OK (text/html)
292	12.593413	10.128.62.58	128.153.5.248	DNS	72	Standard query 0xbadc HTTPS cdn.honey.io
293	12.593500	10.128.62.58	128.153.5.248	DNS	72	Standard query 0x4dc1 A cdn.honey.io
294	12.596814	128.153.5.248	10.128.62.58	DNS	168	Standard query response 0xbadc HTTPS cdn.honey.io SOA ns-cloud-a1.googledomains.com
295	12.598012	128.153.5.248	10.128.62.58	DNS	88	Standard query response 0x4dc1 A cdn.honey.io A 130.211.30.54
297	12.607586	10.128.62.58	34.223.124.45	HTTP	491	GET /favicon.ico HTTP/1.1
322	12.701115	34.223.124.45	10.128.62.58	HTTP	482	HTTP/1.1 200 OK (PNG)

for favicon.io. The packet numbers for the 200 OK were 149, 271, and 322.



The Three Way Handshake

	SYN	SYN,ACK	ACK
Packet Number	42	69	71
Source Port	63990	80	63990
Destination Port	80	63990	80
Time Delta	N/A	95.56 ms	N/A

Internet Protocol V4 Screenshot

The highlighted portion of the Internet Protocol V4 Screenshot, shows that the header spans 20 bytes.

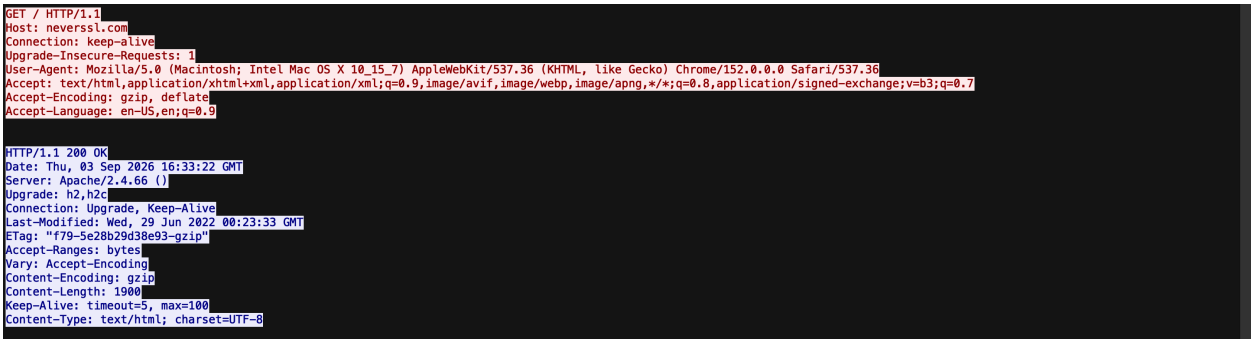
OSI Model

Wireshark Row	OSI Layer	PDU Name	Addresses it Carries
Frame	NONE	_____	_____

Ethernet II	Layer 2 (Data Link)	Frame	MAC address: 3e:d8:af:15:26:54 -> 00:09:0f:09:00:20
Internet Protocol Version 4	Layer 3 (Network)	Packet	Source IP Address & Destination IP Address: 10.128.62.58 -> 34.223.124.45
Transmission Control Protocol	Layer 4 (Transport)	Segment	Source Port & Destination Port 63990 -> 80
Hypertext Transfer Protocol	Layer 7 (Application)	Data / Message	URL : neverssl.com

The reason that layers 5 and 6 do not have their own rows in this OSI model outline is because layer 7 encapsulates these jobs. Functions like session management, data format translation, and encryption are typically built directly into the application protocol rather than handled by separate, standalone network protocols. And layer one does not appear because there is no way that the Wireshark software can see anything in the physical level.

Server Connection



Here is the output from the Follow command of the HTTP connection:

This conversation tells us that the server is running Apache software version 2.4.66

HTTPS

With HTTPS, the information that you can still gather compared to HTTP requests includes the DNS Question Name from DNS lookups, the Source/Destination IP and Source/ Destination Ports during the TCP handshake, and the domain name from the Server Name Indication (SNI) extension. This is because all of these are low-level headers that HTTPS does

not encrypt prior to session establishment. However, once encryption happens, we lose the ability as a passive observer to see application-layer data. Specifically, you cannot read the HTTP Request Method (like GET), the target Request URI, HTTP Headers, or the actual HTTP Payload HTML code sent to the browser. These fields live at the Application layer (Layer 7), which is the primary level HTTPS secures.

CLI

First Ten Lines:

15:02:03.520693 IP 10.128.62.58.23100 > 128.153.5.248.53: 12359+ Type65? www.google-analytics.com. (42)
15:02:03.520932 IP 10.128.62.58.24978 > 128.153.5.248.53: 25900+ A? www.google-analytics.com. (42)
15:02:03.677646 IP 128.153.5.248.53 > 10.128.62.58.24978: 25900 1/0/0 A 142.251.211.78 (58)
15:02:03.677650 IP 128.153.5.248.53 > 10.128.62.58.23100: 12359 0/1/0 (102)
15:02:06.330635 IP 10.128.62.58.65000 > 128.153.5.248.53: 13520+ A? example.com. (29)
15:02:06.367678 IP 128.153.5.248.53 > 10.128.62.58.65000: 13520 2/0/0 A 104.20.23.154, A 172.66.147.243 (61)
15:02:06.372500 IP 10.128.62.58.60145 > 104.20.23.154.80: Flags [SEW], seq 1926797854, win 65535, options [mss 1460,nop,wscale 6,nop,nop,TS val 4213099467 ecr 0,sackOK,eol], length 0
15:02:06.392138 IP 104.20.23.154.80 > 10.128.62.58.60145: Flags [S.E], seq 2294619659, ack 1926797855, win 65535, options [mss 1250,sackOK,TS val 3422386468 ecr 4213099467,nop,wscale 13], length 0
15:02:06.392313 IP 10.128.62.58.60145 > 104.20.23.154.80: Flags [.], ack 1, win 2051, options [nop,nop,TS val 4213099487 ecr 3422386468], length 0
15:02:06.392388 IP 10.128.62.58.60145 > 104.20.23.154.80: Flags [P.], seq 1:75, ack 1, win 2051, options [nop,nop,TS val 4213099487 ecr 3422386468], length 74: HTTP: GET / HTTP/1.1

The Command Line Interface version of tcpdump omits the layer 2 information that Wireshark gives you. It still shows you all of the layer 3 4 and 7 information but in a much more condensed fashion. This makes sense as there is no dropdown or other window to layer information behind when using the command line exclusively.

Reflection

Prior to this lab, I knew that HTTP was not a secure protocol to transfer information, but I did not know to what extent that it was insecure. It was pretty cool to actually see the protections that HTTPS provides. The other thing that I learned from this lab that I did not take away from the lecture portion of the OSI model was how the headers of the information were organized in the packets sent across the network. When the lab instructed us to go into the encapsulation and I saw the headers highlighted in the packet information it made the OSI model seem more relevant and make more sense as to its impact in standardizing internet transmission of information.